Table 1: E1: adaptive recovery over the full campaign of 669,574 configurations (P=3, uniform box, noiseless, 200 samples per 10 s). Wilson 95 % intervals. Median maximal parameter error on recoveries 1.3e-11; median wall time per configuration (all rungs) 3.0 s. Configurations whose worst-case total-internal-reflection margin is negative (6.54 % of the box) are recovered at 93.37 % [93.14, 93.60].

	count	share	95 % CI
recovered at T=10 s	558,919	83.47 %	[83.38, 83.56]
cured at T=20 s	72,134	10.77 %	[10.70, 10.85]
cured at T=40 s	15,145	2.26 %	[2.23, 2.30]
cured at T=80 s	4,974	0.74 %	[0.72, 0.76]
recovered, adaptive	651,172	97.25 %	[97.21, 97.29]
unsolved at $T \leq 80$ s	18,402	2.75 %	[2.71, 2.79]
certificate classes on the unsolved (classes co-occur; share of unsolved)			
lattice relation	9,900	53.80 %	[53.08, 54.52]
weak prism	5,577	30.31 %	[29.65, 30.97]
glitch floor (TIR mask / residual)	7,179	39.01 %	[38.31, 39.72]
close pair	66	0.36 %	[0.28, 0.46]
rank-deficient	1,139	6.19 %	[5.85, 6.55]
basin (spectral stage certified)	1,078	5.86 %	[5.53, 6.21]
no certificate (unexplained)	0	0.00 %	[0.00, 0.02]

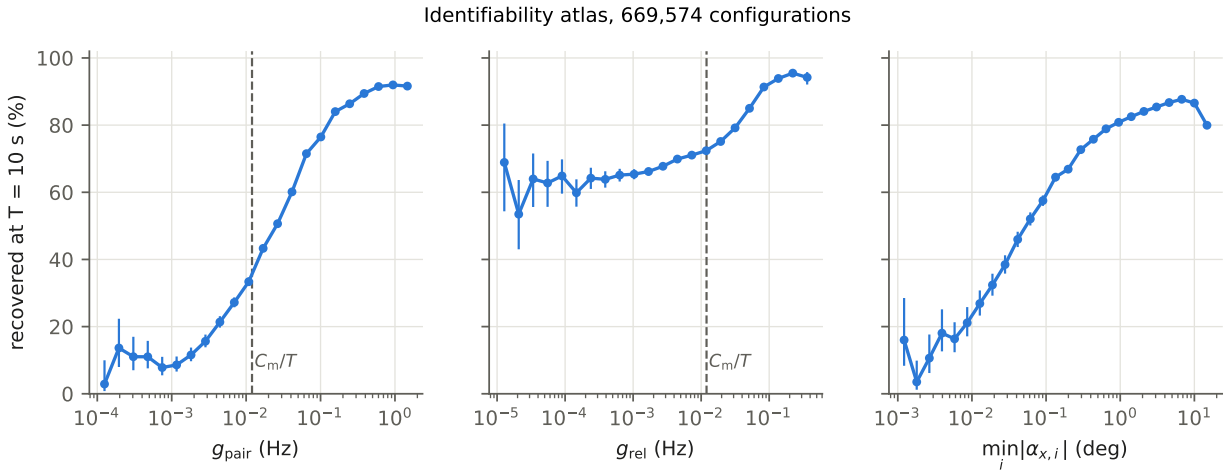


Figure 1: The identifiability atlas: recovery probability at T=10 s against the three margin coordinates (Wilson 95 % bars, bins with ≥20 configurations; 669,574 configurations). Dashed: the merge floor C_m/T = 0.012 Hz. The derived detectability overlay $\alpha_{min}(T)$ is configuration-dependent and is not drawn.

Table 2: E2 (development scale): certified bounds on 121 fresh recovered configurations (seed 20261008, T=10 s), every parameter checked against truth. The 10^7-check campaign version of this table is still pending; this run fixes the protocol and gives a first coverage figure.

	count	share	95 % CI
configurations with all 18 bounds covering	117/121	96.69 %	[91.81, 98.71]
median tightness (bound / true error), median over cases	5.0		

Table 3: E3, frequency-error law only. Population: the 18,402 configurations unsolved at every rung, so the same cases appear at each T (an adaptive-campaign cohort, not the fixed-T design; the pair-gap and detectability laws still need the fixed-T sweep).

law	derived	fitted (bootstrap 95 %)	note
$\sigma(\hat{N}) \propto T^{\gamma}$	$-3/2$	-1.58 [-1.59, -1.56]	T=20–80 only: -1.42
pair-gap inflation	-2	pending	needs fixed-T cohorts
detectability	$-1/2$	pending	needs fixed-T cohorts

Table 4: E4, ladder-censored version: 75,730 configurations failed at T=10 s with a numeric prescription and were later recovered on the ladder. $T_{\min}$ is only known to lie in (previous rung, recovering rung]; the bisection design is still pending. Among the 9,902 configurations unsolved at 80 s with a numeric prescription, 9,594 (96.9 %) were prescribed $T_{\mathrm{req}} > 80$ s, i.e. correctly predicted infeasible within the ladder.

recovered no later than the first rung $\geq T_{\mathrm{req}}$	91.48 % [91.28, 91.68]
$T_{\mathrm{req}}/T_{\mathrm{recovered}}$: p5 / p25 / p50 / p75 / p95	0.40 / 0.80 / 1.25 / 2.60 / 21.90



Figure 2: Certified maximal bound against true maximal error, 121 fresh recoveries (local development-scale run). Coverage = mass above the identity line.

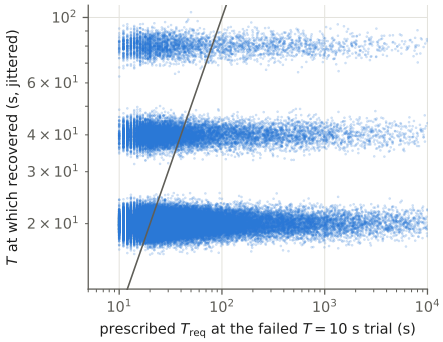


Figure 4: Prescribed T_{req} against the rung that recovered the configuration (vertical jitter for visibility); identity line drawn.

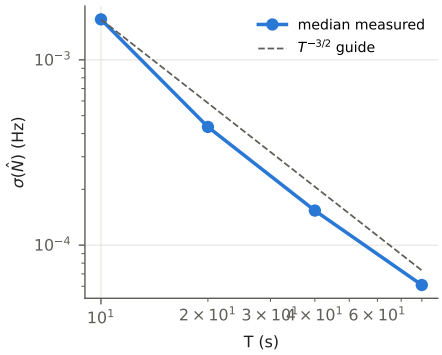


Figure 3: Median certified $\sigma(\hat{N})$ against T for the always-unsolved cohort, with the derived $T^{-3/2}$ guide.